

Domain-Adapted GPS Prediction for Italian Landscapes via Parameter-Efficient Fine-Tuning of GeoCLIP

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Abstract

We present a parameter-efficient approach to fine-tune GeoCLIP, a CLIP-based geo-localization model, for predicting GPS coordinates from photographs taken in Italy. Starting from a pretrained 440M-parameter backbone, we inject Low-Rank Adaptation (LoRA) adapters into all four attention projection matrices of the Vision Transformer encoder, replace GeoCLIP’s gallery-retrieval head with a direct coordinate-regression head, and update only 1.4% of the weights. We train on 111,787 images from the OpenStreetView-5M (OSV5M) dataset filtered to Italy’s borders via a polygon boundary test, using a combined Haversine and L1 regression loss augmented with a land-aware penalty that discourages predictions in the open sea. Because fair and reproducible comparison is central to this work, we evaluate every model, including both baselines and fine-tuned variants, through a single inference-and-metrics pipeline on the fixed OSV5M test set of 1,397 images. Our best model (V3.3 with test-time augmentation) achieves a median error of 110.8 km and 73.6% of predictions within 200 km, compared to 538.3 km / 24.6% for the unmodified GeoCLIP baseline and 245.2 km / 42.8% for GeoCLIP restricted to an Italy-only retrieval gallery. Against this stronger, Italy-constrained baseline our model still cuts median error by 55% (245.2→110.8 km), showing the gain is not an artifact of comparing against an unconstrained global gallery. Macro-region classification accuracy improves from 63.5% to 80.2%. An ablation traced through the same pipeline shows how dataset scale, LoRA capacity, adapter scope, and test-time augmentation each contribute.

1. Introduction

Visual geo-localization, the task of predicting the geographic location depicted in a photograph, has applications in navigation, social-media analysis, forensic verification, and cultural heritage documentation. General-purpose models such as PlaNet [11] and GeoCLIP [9] achieve impressive worldwide coverage, but their performance in spe-

cific countries often lags behind what dedicated domain adaptation can achieve.

Italy is a compelling target for regional geo-localization. Its diverse landscape spans Alpine mountains, Mediterranean coastlines, historic urban centers, and rural agricultural areas, all within a compact bounding box (35.4–47.2°N, 6.6–18.8°E). A practical geo-localization system for Italy could support tourism, cultural heritage research, and automated geo-tagging of social-media photographs.

Our first attempts fine-tuned GeoCLIP by directly unfreezing the last Vision Transformer (ViT) layers. Applying a standard learning rate (10^{-4}) to those layers produced behavior consistent with catastrophic forgetting [6], aggressively overwriting the fine-grained feature detectors acquired during global pretraining. This motivated a shift to parameter-efficient adaptation: we freeze the ViT transformer blocks and inject LoRA adapters [5] whose weight matrices start at zero and grow only to capture Italy-specific signals, while additionally training the visual projection and image-encoder MLP as lightweight domain-adaptation components.

Beyond the model, this paper emphasizes *evaluation consistency*. Comparing a regression model with a retrieval model requires care because GeoCLIP predicts the nearest entry in a fixed gallery of GPS candidates, most of which lie outside Italy, whereas our regression head is structurally confined to Italy. To ensure a fair comparison, we evaluate GeoCLIP both with its full global gallery and with a gallery restricted to Italy, alongside the zero-shot StreetCLIP [3] retriever, and run all of them, plus every fine-tuned checkpoint, through one shared inference-and-metrics pipeline on the identical test set. All reported quantitative results and figures are produced using this shared evaluation pipeline.

Main contributions:

- A parameter-efficient fine-tuning pipeline that converts GeoCLIP’s retrieval design into direct GPS regression, updating only 1.4% of a 442M-parameter backbone.
- A differentiable land-aware penalty loss designed to discourage off-coast predictions along Italy’s extensive coastline.

- A single unified evaluation harness that measures pre-trained baselines (global gallery, Italy-only gallery, StreetCLIP) and every fine-tuned version identically, resolving the inconsistency of comparing a regressor to differently-configured retrievers.
- A version-by-version ablation and region-level analysis on the fixed OSV5M Italy test set, showing an 87% reduction in mean error and a macro-region classification accuracy of 80.2%.

2. Related Work

Image geo-localization. Hays and Efros [4] showed that scene statistics carry a geographic signal strong enough for continent-level retrieval. PlaNet [11] trained a CNN classifier over a discretized S2 cell grid, achieving scalable worldwide localization. Transformer-based approaches [8] leverage the global receptive field of self-attention for scene-level cues. Our work differs in that we target a single country through regression rather than classification, preserving metric GPS coordinates.

GeoCLIP. Vivanco Cepeda *et al.* [9] align CLIP image embeddings with location encodings derived from random Fourier features of GPS coordinates via contrastive pre-training on 100M image-GPS pairs. At inference, images are ranked against a gallery of GPS candidates. This gallery-based design is strong for worldwide localization but is not optimized for dense intra-country coverage: only about 4,448 of its 100K gallery cells fall inside Italy, motivating our regression approach.

Parameter-efficient fine-tuning. LoRA [5] adds low-rank adapter matrices $\mathbf{BA}$ (initialized with $\mathbf{B} = \mathbf{0}$) in parallel with frozen weight matrices, keeping the original embedding space intact. Hu *et al.* showed that rank-8 adapters match full fine-tuning on NLP benchmarks while updating $< 1\%$ of the parameters. The PEFT library [7] provides turnkey LoRA support for Hugging Face transformers.

StreetCLIP. Haas *et al.* [3] present StreetCLIP, a generalized zero-shot learner for open-domain image geolocalization. GPS candidates are encoded as text prompts and the nearest match to the image embedding is returned. On our Italy-constrained gallery (50K OSV5M train points encoded as text), StreetCLIP reaches a median error of 438.8 km on OSV5M Italy, illustrating that general zero-shot geolocalization does not substitute for domain-specific adaptation.

OSV5M. Astruc *et al.* [1] release five million geo-tagged street-view images from OpenStreetMap contributors worldwide with a fixed train/test split. Unlike landmark-centric GLDV2 [10], OSV5M covers everyday roads, intersections, and rural areas, making it substantially better suited for regional geo-localization.

3. Dataset

3.1. OpenStreetView-5M Italy Subset

We filter OSV5M [1] to images inside Italy’s borders using a point-in-polygon test against a Natural Earth Italy boundary (via a prepared `shapely` geometry, buffered 0.02° to keep coastal points), rather than the coarser $35.4\text{--}47.2^\circ\text{N}$, $6.6\text{--}18.8^\circ\text{E}$ bounding box, which is used only as an outer pre-filter and as the regression target range in §4. This excludes points that fall inside the bounding box but outside Italy (e.g., in neighboring countries or the sea), yielding 111,787 training images and 1,397 test images. The test split is the fixed OSV5M partition; it is never used during training or model selection, and it is the sole evaluation set for all quantitative results in this paper. For development, we split the training pool 85/15 (`random_state=42`), stratified by macro-region from V3.2 onward to prevent the Nord-heavy pool from over-representing northern Italy in the validation set.

Regional distribution. We use Italy’s standard statistical macro-regions: Nord (North), Centro (Center), and Sud/Isole (South and Islands). The training pool skews north: Nord 57.6%, Sud/Isole 23.1%, Centro 19.3%. The test set presents the opposite skew: Sud/Isole 734 images (52.5%), Nord 343 (24.6%), Centro 320 (22.9%). This mismatch partially explains why Sud/Isole recall (74.7%) trails Nord (98.3%) in the fine-tuned model (§5.5).

3.2. Legacy GLDV2 Experiments

Two preliminary versions (V1, V2) used an Italy subset of GLDV2 [10] ($\sim 34\text{K}$ landmark images after CLIP-fingerprint deduplication). GLDV2’s heavy bias toward famous monuments gave poor sub-city precision, motivating the switch to OSV5M. We report GLDV2 numbers only briefly for completeness (§5.7); they are on a different test set and are not comparable to the OSV5M results that form the body of this paper.

3.3. IM2GPS3k Italy Subset

For a cross-dataset generalization check (§5.6), we additionally use IM2GPS3k [4], the $\sim 3,000$ -image worldwide test benchmark of Hays and Efros, filtered to Italy with the same polygon boundary test used for OSV5M (§3.1). This yields 110 images, predominantly landmark and tourist-scene photographs, in contrast to OSV5M’s everyday street-level coverage. IM2GPS3k is never used for training or model selection; it serves only as an out-of-distribution evaluation set.

3.4. Preprocessing and Augmentation

For validation and inference, images are resized to 224×224 . During training we apply random crop ($256 \rightarrow 224$), horizontal flip ($p = 0.5$), color jitter

Ver.	LoRA targets	r	Trainable	%
V1	None (2 ViT layers)	–	28.1M	6.4
V2	q, v	8	≈ 2 M	0.5
V3.1	q, v	8	≈ 2 M	0.5
V3.2	q, k, v, out	16	6.1M	1.4
V3.3	q, k, v, out	16	6.1M	1.4

Table 1. Trainable parameter counts. V3.2/V3.3 breakdown: 3.15M LoRA adapters, 0.79M visual projection, 0.98M MLP, 1.19M regressor.

(brightness/contrast ± 0.3 , saturation ± 0.2 , hue ± 0.05), random rotation ($\pm 15^\circ$), and random grayscale ($p = 0.05$). All images are normalized with the OpenAI CLIP mean $[0.481, 0.458, 0.408]$ and standard deviation $[0.269, 0.261, 0.276]$. Augmentations were fixed across all versions to avoid confounding train/test distribution shifts.

4. Method

4.1. From Retrieval to Regression

GeoCLIP [9] consists of a CLIP ViT-Large/Patch14 image encoder and a location encoder that maps GPS coordinates to 512-d vectors. At inference time, it performs nearest-neighbor lookup over a gallery of GPS candidates. We discard the location encoder and the gallery entirely and attach a direct regression head to the image features. Our fine-tuned model performs *no* gallery lookup at any stage; it emits a continuous (ϕ, λ) pair. In our experiments, the gallery is used only by the pretrained retrieval baselines described in §5.1 and not by our regression model.

4.2. Adaptation Strategy

The ViT transformer blocks are frozen throughout training. LoRA adapters are injected into the attention projection matrices of every ViT encoder layer, allowing the attention mechanism to adapt without modifying the frozen weights. The adapter update is

$$\Delta \mathbf{W} = \frac{\alpha}{r} \mathbf{B} \mathbf{A}, \quad \mathbf{A} \in \mathbb{R}^{r \times d}, \mathbf{B} \in \mathbb{R}^{d \times r}, \mathbf{B} = \mathbf{0} \text{ at init,} \quad (1)$$

where r is the rank and α the scaling factor. Additionally, the visual projection (512×768) and image-encoder MLP are trained as a lightweight domain-adaptation bottleneck between the frozen ViT features and the regression head. In V3.2/V3.3 the LoRA targets expand from $\{q, v\}$ to all four projections $\{q, k, v, \text{out}\}$ and the rank doubles from 8 to 16. Table 1 summarizes trainable parameter counts.

4.3. Regression Head

The 512-d CLIP pooled feature passes through $512 \rightarrow 1024 \rightarrow 512 \rightarrow 256 \rightarrow 2$, with BatchNorm + GELU +

Dropout ($p=0.3, 0.2, 0.1$) between layers and a final Sigmoid. The output $\hat{y} \in [0, 1]^2$ maps to coordinates by

$$\hat{\phi} = \hat{y}_0 \cdot (\phi_{\max} - \phi_{\min}) + \phi_{\min}, \quad (2)$$

$$\hat{\lambda} = \hat{y}_1 \cdot (\lambda_{\max} - \lambda_{\min}) + \lambda_{\min}, \quad (3)$$

with $[\phi_{\min}, \phi_{\max}] = [35.4^\circ, 47.2^\circ]$ and $[\lambda_{\min}, \lambda_{\max}] = [6.6^\circ, 18.8^\circ]$. The Sigmoid confines every prediction to Italy’s bounding box by construction.

4.4. Loss Function

Base loss. We combine an L1 term (normalized coordinates) with a Haversine distance term (km, divided by 1000 for stability):

$$\mathcal{L}_{\text{base}} = 0.05 \mathcal{L}_{L1}(\hat{y}, y) + 0.95 \mathcal{L}_{\text{Hav}}(\hat{p}, p). \quad (4)$$

The 95:5 weighting prioritizes the geographically meaningful metric while the L1 term provides a smooth gradient near the optimum.

Land-aware penalty (V3.1–V3.3). Italy’s long coastline makes many ambiguous scenes (harbors, beaches) prone to offshore predictions. We add a differentiable penalty that is zero on land or within $d_{\text{buf}} = 25$ km of the coast and ramps linearly to 1.0 at $d_{\text{scale}} = 150$ km offshore:

$$\mathcal{L}_{\text{land}}(\hat{p}) = \frac{1}{B} \sum_{i=1}^B \text{clip}\left(\frac{d_i - d_{\text{buf}}}{d_{\text{scale}}}, 0, 1\right), \quad (5)$$

where d_i is the distance from the i -th prediction to the nearest land pixel. The grid is precomputed at 300×300 over Italy using `global_land_mask` and `scipy`’s distance transform, and sampled differentially via `F.grid_sample`. The combined loss is $\mathcal{L} = \mathcal{L}_{\text{base}} + 0.20 \mathcal{L}_{\text{land}}$.

4.5. Test-Time Augmentation (V3.3)

At inference we average GPS predictions over six deterministic views: standard resize; horizontal flip; center-crop from 256 px; flipped center-crop; a fixed brightness/contrast increase; and a fixed $+10^\circ$ rotation. Averaging coordinates reduces variance with no retraining.

5. Experiments

5.1. Unified Evaluation Protocol

All models are evaluated by one shared harness that loads the fixed OSV5M test set, runs inference, and computes Haversine error and threshold accuracies identically. We compare our fine-tuned checkpoints against four reference settings. The first is **GeoCLIP (global gallery)**, the model as shipped, retrieving over all ≈ 100 K GPS cells. The second is **GeoCLIP (Italy gallery)**, the same model with

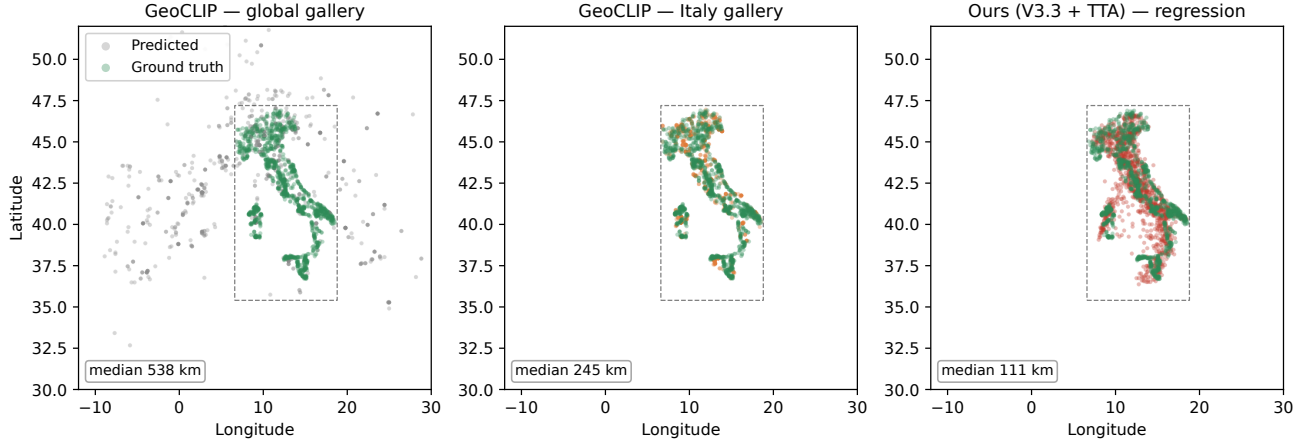


Figure 1. Predicted (colored) vs. ground-truth (green) GPS coordinates on the OSV5M Italy test set (1,397 images); dashed box is Italy’s bounding box. **Left:** pretrained GeoCLIP with its full global gallery scatters predictions across Europe and beyond. **Center:** restricting GeoCLIP’s gallery to Italy forces in-country answers but remains coarse. **Right:** our regression model (no gallery) clusters tightly on the true locations. Median errors are annotated per panel.

Ver.	BS	Eff. BS	LR	WD	Ep.	Sched.
V1	16	16	1e-4	1e-4	30	CosAnn
V2	16	16	1e-4	1e-4	30	CosAnn
V3.1	32	32	1e-4	1e-4	15	CosAnn
V3.2	32	128	1e-4	1e-4	15	Warm+Cos
V3.3	32	128	5e-5	1e-4	15	Warm+Cos

Table 2. Training hyperparameters. Eff. BS = effective batch size after gradient accumulation ($\times 4$). Warm+Cos = 500-step linear warmup then cosine annealing to $\eta_{\min} = 10^{-6}$.

its gallery restricted to the 4,448 cells inside the Italy polygon; this is a fair baseline, since it receives the same “it is in Italy” prior that our regressor has by construction. The third is **StreetCLIP** [3], which performs zero-shot retrieval over 50K text-encoded OSV5M train locations. The fourth setting is our own regression models. Only the retrieval baselines use a gallery; our regression model does not.

5.2. Training Details

All models run on Google Colab T4 GPUs with automatic mixed precision. Table 2 lists key hyperparameters. V3.3 additionally applies `torch.compile` and a larger `DataLoader prefetch_factor` to hide Google Drive I/O latency; its best checkpoint was reached at epoch 10 (val loss 0.1054).

5.3. Main Results and Ablation

Table 3 reports every model through the shared pipeline on the same 1,397-image test set, so all rows are directly comparable. Figure 2 visualizes the median-error progression.

Baselines. Restricting GeoCLIP’s gallery to Italy reduces

Model	Mean	Median	≤ 25	≤ 200
GeoCLIP (global gallery) [9]	1290.5	538.3	3.9	24.6
StreetCLIP (Italy) [3]	467.9	438.8	0.9	16.6
GeoCLIP (Italy gallery)	301.7	245.2	6.8	42.8
V3.1 (LoRA $r=8$)	195.2	124.2	7.4	66.1
V3.2 (LoRA $r=16$)	186.2	122.2	7.1	68.3
V3.3 (no TTA)	183.9	119.7	8.8	68.9
V3.3 + TTA	168.0	110.8	10.6	73.6

Table 3. Localization error (km) and threshold accuracy (%) on the OSV5M Italy test set, all measured by the same pipeline. Top block: pretrained retrieval baselines. Bottom block: our fine-tuned regression models. GeoCLIP (Italy gallery) restricts retrieval to 4,448/100K cells inside Italy.

its median error from 538.3 km to 245.2 km, indicating that a substantial portion of the unconstrained baseline error comes from predictions outside Italy. We therefore treat the Italy-gallery GeoCLIP model as the stronger baseline. Figure 1 illustrates this behavior: the global gallery produces predictions across Europe, while the Italy-restricted gallery keeps predictions within the country but remains spatially coarse. StreetCLIP, despite being CLIP fine-tuned for geolocalization, reaches a median error of 438.8 km, higher than the Italy-gallery GeoCLIP.

Our models. All fine-tuned versions outperform the evaluated baselines on median error and threshold accuracy. Relative to the global baseline, V3.3+TTA cuts mean error by 87% (1290.5 $\rightarrow$ 168.0 km) and median error by 79% (538.3 $\rightarrow$ 110.8 km). Relative to the stronger Italy-gallery baseline it still cuts median error by 55% (245.2 $\rightarrow$ 110.8 km), so the improvement is not an artifact of an unfair global comparison. Within the fine-tuned mod-

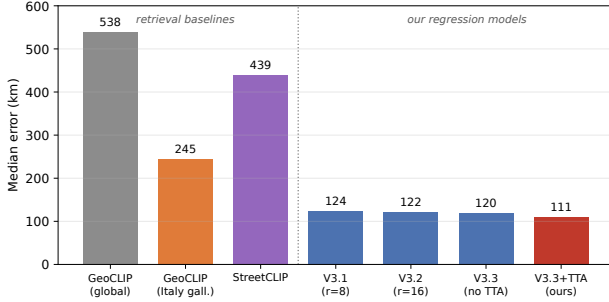


Figure 2. Median localization error on the OSV5M test set, all measured by the same pipeline. The three retrieval baselines (left) have higher median error than all fine-tuned regression models (right); each fine-tuning step adds a consistent improvement, with TTA reducing the median error to 111 km.

Model	1	25	50	100	200	500
GeoCLIP (global)	0.1	3.9	6.8	14.0	24.6	47.2
GeoCLIP (Italy)	0.1	6.8	13.3	26.8	42.8	77.5
StreetCLIP	0.0	0.9	2.4	5.3	16.6	60.0
V3.3+TTA (ours)	0.1	10.6	24.2	45.7	73.6	94.3

Table 4. Accuracy (%) at distance thresholds (km) on the OSV5M test set.

els, increasing the LoRA rank and adapter scope reduces the median error from 124.2 km to 122.2 km. The refined learning-rate schedule used in V3.3 without TTA further reduces it to 119.7 km, and six-view TTA brings it down to 110.8 km (a 7.4% reduction) without additional training. The small deltas between adjacent fine-tuned versions (e.g., V3.1→V3.2) are not validated with per-image confidence intervals and should be read as trends rather than statistically significant differences; the TTA improvement and the gap relative to the evaluated baselines are substantially larger than the adjacent fine-tuning differences.

On the land penalty. The land-aware penalty is included in all OSV5M-trained models (V3.1–V3.3). Because it was introduced together with the move from GLDv2 to OSV5M, its individual quantitative contribution to aggregate error is not isolated in our ablation; we adopt it as a qualitative design choice that keeps predictions on land, and attribute the large 124.2 km drop at this stage to the dataset switch and LoRA adaptation.

5.4. Threshold Accuracy and Error Distribution

Table 4 gives fine-grained threshold accuracy for the three baselines and our final model, and Figure 3 shows the full error distribution and accuracy-at-distance curves.

City-level precision (≤ 25 km) improves from 3.9% (global) and 6.8% (Italy gallery) to 10.6%, and province-level (≤ 100 km) from 14.0%/26.8% to 45.7%. The 90th-percentile error drops from 1931.9 km (global) to 425.5 km (ours). The persistent gap at the 1 km street-level threshold

Region	GeoCLIP (global)			V3.3 + TTA (ours)		
	P	R	F1	P	R	F1
Nord	0.58	0.74	0.65	0.78	0.98	0.87
Centro	0.41	0.49	0.45	0.60	0.74	0.66
Sud/Isole	0.82	0.65	0.73	0.96	0.75	0.84
Accuracy	0.635			0.802		

Table 5. Macro-region classification on the OSV5M test set (P/R/F1). Test-set support: Sud/Isole $n=734$, Nord $n=343$, Centro $n=320$.

suggests that the 512-d globally pooled CLIP representation may not preserve sufficient fine-grained spatial information for precise localization, discussed further in §6.

5.5. Macro-Region Classification

Assigning each prediction to a macro-region (Nord: $\text{lat} > 44.0$; Centro: $41.5 < \text{lat} \leq 44.0$; Sud/Isole otherwise), overall region accuracy rises from 63.5% (global baseline) and 66.2% (Italy gallery) to 80.2% for V3.3+TTA. These fixed latitude cutoffs are only a coarse proxy for Italy’s administrative and cultural regions. Sardinia, for instance, spans a wide latitude range, so assignments near the thresholds may be imprecise. Table 5 gives the per-region breakdown against the global baseline.

Nord achieves very high recall (98.3%), reflecting the model’s stronger northern training signal, with a per-region median error of 71.7 km. Sud/Isole has the highest median error, at 159.2 km; this region represents more than half of the test set but less than a quarter of the training images. Centro remains the weakest by F1 (0.66): its intermediate latitude overlaps both neighbors, and the model tends to push ambiguous cases toward the better-represented Nord.

5.6. Cross-Dataset Check: IM2GPS3k

On the landmark-heavy IM2GPS3k Italy subset (§3.3) our model obtains a median error of 106.5 km and a mean error of 152.3 km, but here the retrieval baselines perform better at fine spatial scales: GeoCLIP with an Italy gallery reaches a median error of 21.4 km because many famous monuments are close to existing gallery cells, and even the global gallery reaches a median error of 32.1 km, although its mean error increases to 762.4 km due to large-error cases. This suggests a distribution-dependent behavior: retrieval performs better on iconic landmarks that are close to gallery locations, whereas our model is optimized for ordinary street-level scenes. The best method depends on the image distribution, and our target is the everyday-scene regime of OSV5M.

Uncertainty. Following Gal and Ghahramani [2], MC Dropout (50 stochastic passes with BatchNorm frozen) yields lower prediction spread on distinctive urban scenes

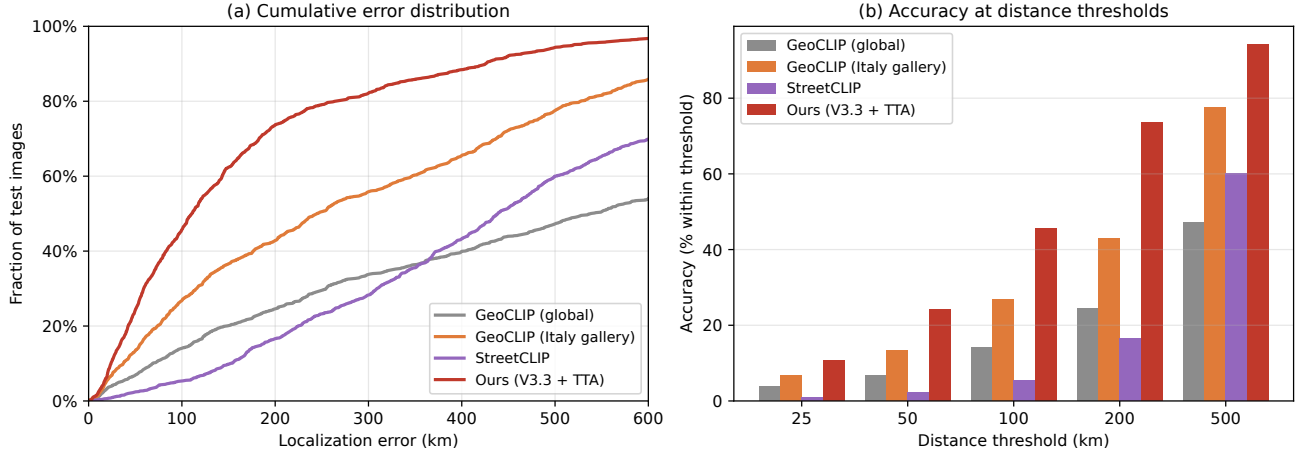


Figure 3. (a) Cumulative error distribution and (b) accuracy at distance thresholds on the OSV5M Italy test set, for the three retrieval baselines and our V3.3+TTA model. Our model dominates across the whole error range; the largest absolute gains over the Italy-gallery baseline appear in the 25–200 km band.

than on featureless coastal or agricultural ones, a promising confidence signal, though we have not validated that the spread correlates with true error, which would be required before deployment.

5.7. GLDv2 Results for Reference

For completeness, the two GLDv2-trained versions reached a median error of 168.0 km (V1) and 124.1 km (V2) on the GLDv2 Italy test split, versus 216.0 km and 235.0 km for their respective GeoCLIP baselines. These numbers are on a different test set from a landmark-biased dataset and are *not* comparable to the OSV5M results above; they are reported only to document the progression that motivated the switch to OSV5M.

6. Conclusion

We adapted GeoCLIP to Italy-specific GPS regression by updating only 1.4% of its parameters via LoRA on all four ViT-L/14 attention projections, replacing its gallery-retrieval head with direct coordinate regression. Evaluated through a single consistent pipeline on the fixed OSV5M test set, our best model (V3.3+TTA) reduces mean error by 87% (1290.5→168.0 km) and median error by 79% (538.3→110.8 km) versus the pretrained baseline, and still cuts median error by 55% against the stronger Italy-gallery baseline, while lifting macro-region accuracy from 63.5% to 80.2%. A differentiable land-aware penalty is included to discourage open-water predictions, although its individual contribution is not isolated in the ablation, and six-view test-time augmentation contributes a final variance reduction with no retraining.

Limitations and future work. City-level precision remains limited, with only 10.6% of predictions falling within

25 km and 0.1% within 1 km. The model is therefore better suited to coarse regional localization than to precise GPS estimation. The IM2GPS3k evaluation indicates that retrieval-based methods remain stronger on iconic landmark images. Promising extensions include (i) hierarchical regression with a coarse region classifier feeding a fine-grained head; (ii) rebalancing the Nord-heavy training set via stratified oversampling or loss reweighting; (iii) patch-level rather than globally-pooled CLIP features to recover the spatial detail sub-city precision needs; and (iv) auxiliary metadata (orientation, season) as additional regression inputs.

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